

1. (a) $\dim \text{Col}(A) = 6$ (b) $\dim \text{Col}(A^T) = 6$ (c) $\dim \text{Row}(A) = 6$
 (d) $\dim \text{Row}(A^T) = 6$ (e) $\dim \text{Nul}(A) = 2$ (f) $\dim \text{Nul}(A^T) = 5$

2. (a) False since Δ free variables
 (b) False there is not a pivot in every row.
 (c) False There are free variables

3. $A = \begin{bmatrix} -2 & 4 \\ 3 & -1 \end{bmatrix}$. $\det(A - \lambda I) = (-2 - \lambda)(-1 - \lambda) - 12 = \lambda^2 + 3\lambda - 10$

(a) Characteristic equation is $\lambda^2 + 3\lambda - 10 = 0$

(b) $(\lambda + 5)(\lambda - 2) = 0$ $\lambda = -5$ or $\lambda = 2$

4. Find basis for each eigenspace $\lambda = 2 \Rightarrow A - 2I = \begin{bmatrix} -2 & 0 & 1 \\ 0 & 0 & 0 \\ 4 & 0 & 2 \end{bmatrix} \sim \begin{bmatrix} -2 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

so $x_1 = -\frac{1}{2}x_3$
 $x_2 = x_2$
 $x_3 = x_3$
 $= x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix}$ e-vectors for $\lambda = 2$ are $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix}$

For $\lambda = -2$ we get $A + 2I = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 4 & 0 \\ 4 & 0 & 2 \end{bmatrix} \sim \begin{bmatrix} 2 & 0 & 1 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ so $x_1 = -\frac{1}{2}x_3$
 $x_2 = 0$
 $x_3 = x_3$

$\therefore$ e-vector for $\lambda = -2$ is $\begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix}$

A is diagonalizable and $P = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 2 & -2 \end{bmatrix}$

5. $\vec{0} \in ES$ since $A \cdot \vec{0} = \vec{0} = \lambda \cdot \vec{0} \quad \therefore ES$ is not empty,

if $\vec{x}$ and $\vec{y} \in ES$ then $A\vec{x} = \lambda\vec{x}$ and $A\vec{y} = \lambda\vec{y}$

$\therefore A(\vec{x} + \vec{y}) = A\vec{x} + A\vec{y} = \lambda\vec{x} + \lambda\vec{y} = \lambda(\vec{x} + \vec{y}) \quad \therefore \vec{x} + \vec{y} \in ES$
 and ES is closed under vector addition.

Let c be any scalar and $\vec{x} \in ES$.

Then $A(c\vec{x}) = c \cdot A\vec{x} = c \cdot \lambda\vec{x} = \lambda \cdot (c\vec{x}) \quad \therefore c\vec{x} \in ES$

and ES is closed under scalar mult.

Since ES is nonempty and closed under addition and scalar mult, it is a subspace.

6. $A \sim B \Rightarrow A = PBP^{-1}$. B is diagonalizable $\Rightarrow B = QDQ^{-1}$ for some Q and diagonal matrix D .
 $\therefore A = P(QDQ^{-1})P^{-1} = (PQ) \cdot D \cdot (PQ)^{-1}$
 $\therefore A$ is diagonalizable.

$$7. \textcircled{a} M = \begin{matrix} & \begin{matrix} S & NS \end{matrix} \\ \begin{matrix} S \\ NS \end{matrix} & \begin{bmatrix} .65 & .15 \\ .35 & .85 \end{bmatrix} \end{matrix}$$

$$\textcircled{b} \begin{bmatrix} .65 & .15 \\ .35 & .85 \end{bmatrix} \begin{bmatrix} .4 \\ .6 \end{bmatrix} = \begin{bmatrix} .35 \\ .65 \end{bmatrix} \quad \begin{matrix} 65\% \text{ non-smokers} \\ \text{in 2012} \end{matrix}$$

~~© Find e-vector for $M-I = \begin{bmatrix} -.35 & .15 \\ .35 & -.15 \end{bmatrix} \sim \begin{bmatrix} .35 & -.15 \\ 0 & 0 \end{bmatrix} \Rightarrow .35x_1 = .15x_2$~~
~~e-vector = $\begin{bmatrix} .15 \\ .35 \end{bmatrix} = \begin{bmatrix} 3/7 \\ 1 \end{bmatrix}$ or $\begin{bmatrix} 3 \\ 7 \end{bmatrix}$ or $\begin{bmatrix} .30 \\ .70 \end{bmatrix}$ expect 30% to be smokers, in the long run.~~

8. $\begin{bmatrix} 1 & 1 & 2 \\ 0 & -1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$ has 3 linearly independent columns so the 3 polynomials are linearly independent in the 3 dimensional space $\mathbb{P}_2$
 i.e. the vectors form a basis for $\mathbb{P}_2$

$$\textcircled{b} \begin{bmatrix} 11 \\ -23 \\ -5 \end{bmatrix} \quad \textcircled{c} \begin{bmatrix} 1 & 1 & 2 & 11 \\ 0 & -1 & -4 & -23 \\ 0 & 0 & 1 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -22 \\ 0 & 1 & 0 & 43 \\ 0 & 0 & 1 & -5 \end{bmatrix} \text{ so } \begin{bmatrix} 11-28t-5t^2 \\ 43 \\ -5 \end{bmatrix}_B$$

$$9. \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 0 & 1 & 0 & 0 & 0 \\ 3 & 1 & 1 & 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -1 & -1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{array} \right] \text{ so } e_1 = -\frac{1}{2} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$A\vec{e}_1 = A\left(-\frac{1}{2} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right) = -\frac{1}{2} \cdot 1 \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} + \frac{1}{2}(-1) \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{2} - \frac{1}{2} + 2 \\ -1 + 0 + 2 \\ -\frac{3}{2} - \frac{1}{2} + 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$